

1954 to December 2011. The correlation is 0.89, lower than the almost perfect correlation for the Fed funds rate and T-bills but still very high. Treasury yields have the same triangle pattern as the short end of the yield curve, but there are some periods of pronounced differences between the short and long ends of the curve. Below we discuss what drives this *term spread*, but notice that the whole yield curve displays a strong degree of commonality with the Fed funds rate.

The strong co-movement of the Fed funds rate and the entire Treasury yield curve is captured by a *level factor*. The level factor simply describes the tendency of all bond yields to move up and down together, largely in line with the Fed funds rate. Level factor movements constitute the vast majority—above 90%—of movements across all maturities in the yield curve (or the *term structure of interest rates*).⁶ The exposure to level is known as *duration*. This is analogous to beta in equity markets, which measures exposure to the market equity factor (see chapter 6). Duration is the beta of bonds with respect to level shifts of interest rates. Other factors influencing bond returns are *slope* or the *term spread*, which describes the movements of the long end of the yield curve relative to the short end, and *curvature*, which allows the middle of the yield curve to move independently of the level and term spread. But the most important factor is level.

Factors are extremely important in fixed income. It is instructive to compare fixed income with equities in this regard. In equities, idiosyncratic risk is large, constituting 80% to 90% of return variance, and systematic factor risk is small (10% to 20%). Intuitively, each company is different—firms have different managers, plants, employees, and production technologies even when they operate in the same sector. In fixed income, the idiosyncratic risk component is quite small—often less than 1% for risk-free Treasuries. Systematic factor risk, especially level, dominates.

Consider a five-year Treasury bond. This bond is highly correlated (approximately 98%) with the six-year bond, which is highly correlated (approximately 98%) with the seven-year bond. The risk-free cash flow streams of these bonds are fixed, as we assume that the Treasury will pay. Thus, the only uncertainty is the rate at which the cash flows are discounted. The seven-year bond is subject to the same discount rate as a one-year bond, plus a rate prevailing between years one and seven (this is called the *forward rate*). Similarly, the six-year bond is subject to the short-term one-year discount rate, plus a rate prevailing between years one and six. Thus, whatever affects the short-term rate gets propagated through the yield curve—each long-term bond moves in response to the one-year rate as well as shocks that affect longer maturities. The short, one-period rate affects *all* bond prices. This is why the level factor, or the level shift, is so important in fixed income.

Idiosyncratic risk does increase as you move through the risk spectrum of fixed income from sovereigns, to agencies, to investment grade corporate bonds, and

⁶ See, among others, Litterman and Scheinkman (1991).